
ShopNow Purchase Propensity Report

Introduction

This project builds a customer propensity model to predict which ShopNow customers are likely to purchase in the next 30 days and to quantify the dollar value of different modeling choices. The analysis uses a labeled customer-level dataset with demographics, recent purchase behavior, engagement metrics, product category mix, and a binary target.

1. Modeling Approach

Modeling approach describes how the data was prepared, which models were trained to predict 30-day purchase propensity and the comparison of their business value.

1.1 Data Description

Each row represents a unique customer with fields for age, gender, income bucket, orders in the last 12 months, days since last purchase, average order value, website visits in the last 30 days, email opens in the last 90 days, top spending category, category share variables, discount exposure, and the target flag for purchase in the next 30 days.

1.2 EDA Highlights

Below are the EDA findings that affected downstream modeling choices, such as class imbalance, skewed RFM variables, and key behavioral drivers.

- **The target is imbalanced, with roughly 23% purchasers and 77% non-purchasers, which motivates stratified sampling and class-imbalance handling in the models**
- Orders, recency, and average order value are skewed with a few extreme customers
- Website visits and email opens are moderate-range count variables
- Purchasers generally have more orders, more recent activity (lower recency days), slightly higher spend, and higher engagement

1.3 Feature Engineering and Preparation

Below are the key transformations applied before modeling

- Dropped non-predictive identifiers (customer_id) and free-text feedback

- Winsorized the most skewed RFM variables to reduce outlier influence
- Scaled core numeric features for use in regularized linear models
- Categorical variables (gender, income bucket, top category) are one-hot encoded

Class imbalance is handled via class **weights or scale_pos_weight**, and a **stratified 80/20 train-test split** preserves the overall purchase rate for fair evaluation.

1.4 Models Considered

Four supervised classification models are trained on the same prepared feature set:

1. RidgeClassifier (L2 regularization)
2. L1-penalized logistic regression (Lasso)
3. Random Forest
4. XGBoost

Ridge and Lasso provide stable, interpretable baselines under multicollinearity, while Random Forest and XGBoost capture non-linearities and interactions, with all models configured to account for class imbalance so that purchasers are not overwhelmed by the majority non-purchaser class.

2. Evaluation

Goal: compare models both on standard classification metrics and on expected dollar value, so the final choice aligns with business profit rather than just accuracy.

Models were evaluated using 5-fold stratified cross-validation on the training set, followed by final performance assessment on a held-out 20% test set.

2.1 Evaluation Metrics

Used accuracy, precision, recall, and F1 for the purchase class (1) to see how well each model detects likely buyers in an imbalanced setting.

- Recall and F1 for class 1 are emphasized because missing a true purchaser (FN) is more costly than incorrectly targeting a non-purchaser (FP) under the business payoff.

2.2 Profit Framework

Defined a payoff table that assigns +15 to TP, -5 to FP, 0 to TN, and -10 to FN, reflecting revenue from correctly targeted purchasers and the cost of incentives or missed sales.

For each model and threshold, I converted the confusion matrix into total dollar value using this payoff, making model comparison directly tied to expected marketing impact.

2.3 Threshold Tuning

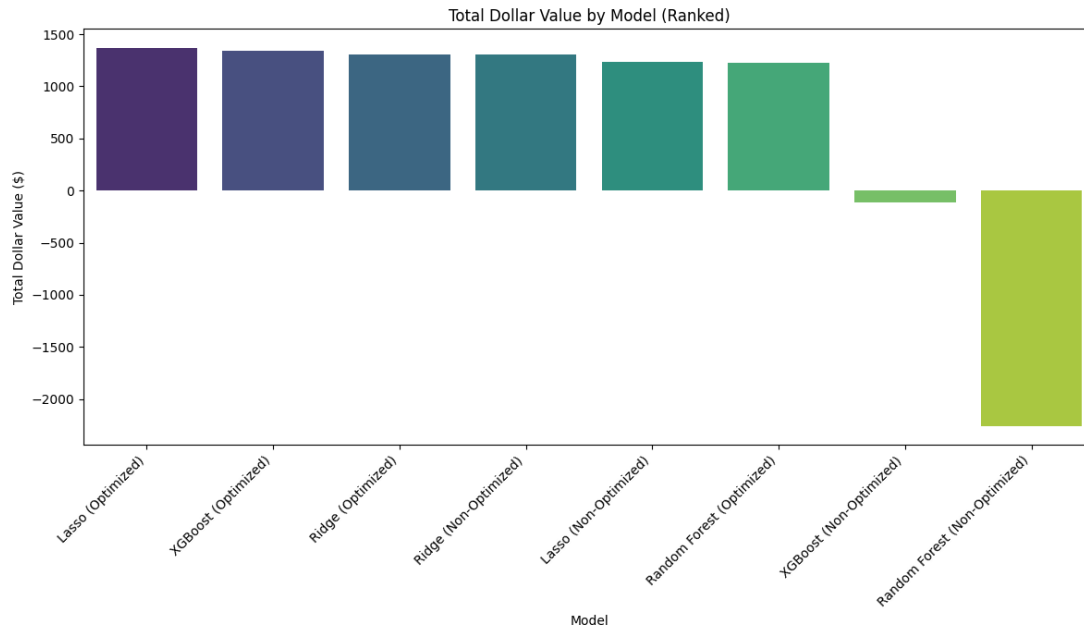
Instead of always using the default 0.5 cutoff, I swept a grid of thresholds over model scores (probabilities for Lasso/RF/XGBoost, decision scores for Ridge) to **find the threshold that maximizes dollar value on the test set**.

This step explicitly trades off more TPs (and some extra FPs) against fewer FNs, aligning the classification boundary with the asymmetric business costs and improving profit for Lasso, RF, and XGBoost.

2.4 Model Comparison

For each model (baseline and optimized), I reported accuracy, precision, recall, and F1 on the test set alongside total dollar value, using the same test split for a fair comparison.

Sr. No.	Model	Accuracy	Precision	Recall	F1-Score	Total Dollar Value
1	Lasso (Optimized)	61%	36%	93%	52%	\$1,370
2	XGBoost (Optimized)	57%	34%	97%	51%	\$1,340
3	Ridge (Non-Optimized)	58%	35%	95%	51%	\$1,305
4	Ridge (Optimized)	58%	35%	95%	51%	\$1,305
5	Lasso (Non-Optimized)	64%	37%	88%	52%	\$1,235
6	Random Forest (Optimized)	61%	36%	91%	51%	\$1,230
7	XGBoost (Non-Optimized)	69%	38%	58%	45%	-\$110
8	Random Forest (Non-Optimized)	78%	52%	8%	14%	-\$2,260



Selected optimized Lasso as the final model because it delivers the highest dollar value while maintaining competitive recall and F1, showing that a regularized linear model with cost-based threshold tuning outperforms more complex trees for this task.

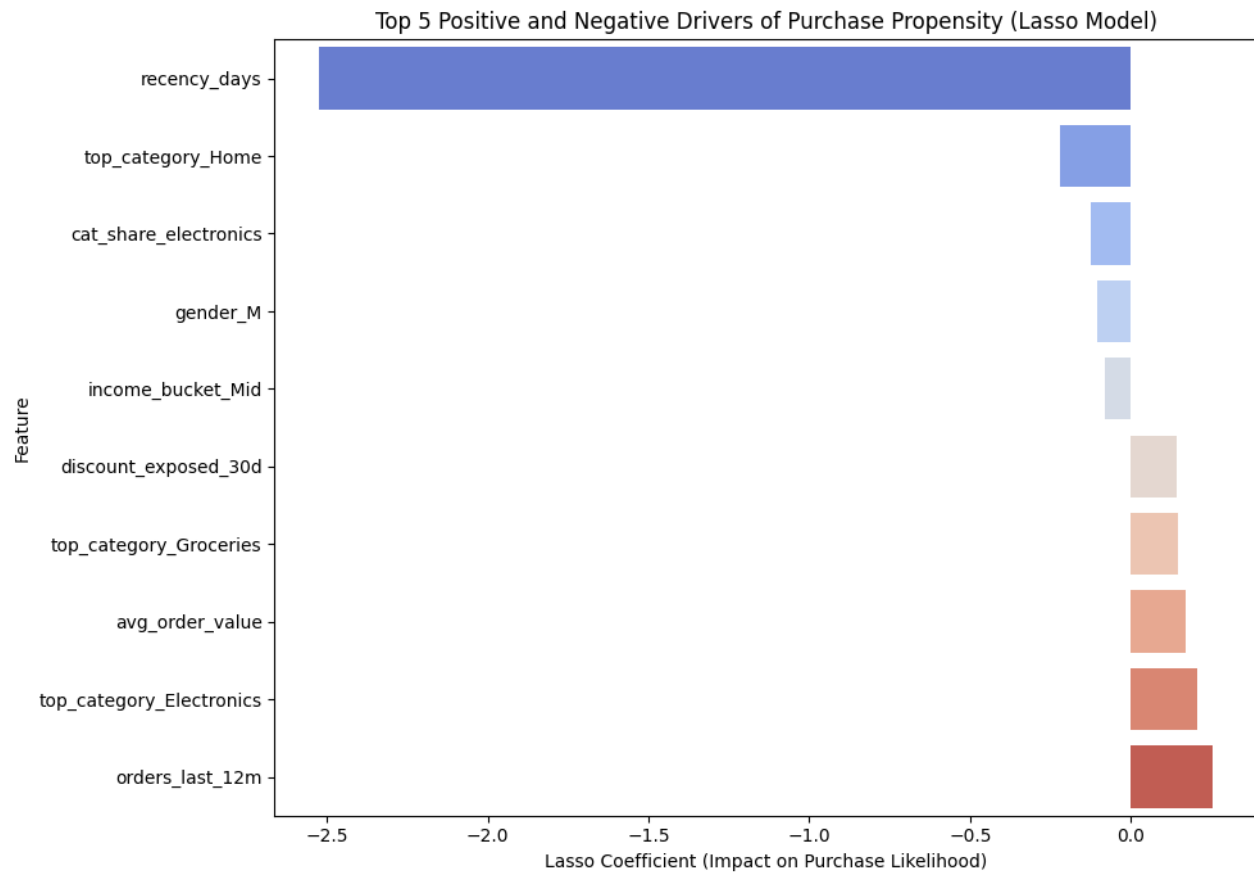
3. Insights

The analysis reveals which customer behaviors most strongly predict short-term purchase and shows that business-aligned threshold tuning can matter more than using the most complex model.

3.1 Behavioral Drivers of Purchase

Lasso coefficients show that the strongest positive drivers of purchase propensity are:

1. lower recency (more recent activity)
2. higher orders in the last 12 months
3. higher average order value



Higher website visits and email opens, as well as higher spend share in certain categories (e.g., Electronics, Groceries, Home), further increase the likelihood of purchase, while some other categories or patterns contribute negatively.

3.2 Effects of Engagement and Discounts

Customers exposed to discounts in the last 30 days and those with higher digital engagement (site visits and email opens) have meaningfully higher predicted purchase probabilities, indicating that marketing touchpoints are effective when combined with existing activity.

However, discounts alone are not sufficient; their impact is strongest among already engaged customers, suggesting that promotions should be layered on top of strong behavioral signals rather than used broadly.

3.3 Modeling Insights

Regularized linear models, especially Lasso, capture most of the signal in a stable and interpretable way and outperform more complex tree models once profit-based threshold tuning is applied.

Comparing accuracy to dollar value highlights that models with slightly lower accuracy but better recall and fewer costly false negatives can generate higher profit, underscoring the importance of evaluating models in terms of business outcomes rather than metrics alone.

4. Recommendations

Use the optimized Lasso model and its profit-maximizing threshold as the core targeting engine for near-term purchase campaigns.

4.1 Targeting Strategy

Prioritize customers above the optimized Lasso threshold as the primary audience for next-30-day promotional outreach, since they generate the highest expected dollar value per contact.

Within that group, further rank by predicted probability to focus the most expensive or scarce interventions (e.g., higher discounts, personalized calls) on the very top deciles.

4.2 Campaign Design

Combine targeted discounts with digital engagement: emphasize promotions via email and site banners to customers who already show recent visits, orders, or opens, where the model indicates the largest uplift.

For medium-propensity customers (just below the threshold), test lighter-touch nudges such as reminders or content emails rather than deep discounts, and monitor whether they move above the threshold over time.

4.3 Measurement and Next Steps

Implement an A/B test where one group is targeted using the Lasso-based ranking and threshold, and a control group uses a simple heuristic (e.g., recent orders only), then compare incremental revenue and realized TP/FP/FN counts.

As next steps, consider incorporating feedback text via sentiment or topic flags, refreshing the model on new cohorts.